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Sem :- II

Project Title :- Time Series Analysis of Two Major Indian IT Companies During April,2021-March,2025

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Introduction

The share market, also known as stock market, is a platform where shares of publicly traded companies are bought and sold. It allows companies to raise capital by issuing shares to investors, who in turn gain partial ownership of the company. The main components of the share market are stock exchanges, brokerage firms and individual and individual investors. The share market plays a crucial role in economy by facilitating investment, enabling wealth creation, and providing companies with access to funding for growth and development. The share market plays a significant role in India's economy. Some components of share market are discussed below:

- Shares (Stocks): Shares represent ownership in a company. When you buy shares of a company, you become a shareholder, which means you own a portion of that company.

- Marketplaces: Shares are traded on stock exchanges such as the New York Stock Exchange (NYSE), NASDAQ, London Stock Exchange (LSE), Bombay stock exchange(BSE), National stock exchange(NSE) etc. These exchanges provide the infrastructure and framework for trading securities.

- Investors: Investors buy shares with the expectation that the company's value will increase over time, allowing them to profit either through dividends paid by the company or by selling their shares at a higher price than they bought them for.

Now, analyzing data from the share market is crucial for making informed investment decisions and planning for the future due to several key reasons:

1. **Identifying Trends and Patterns:** Market data analysis helps investors identify trends in stock prices, trading volumes, and market movements. By recognizing patterns, such as price trends or trading volumes, investors can make predictions about future market movements and adjust their investment strategies accordingly.
2. **Risk Management:** Analyzing historical and current market data allows investors to assess risks associated with specific stocks or the overall market. This includes understanding volatility, correlation between different assets, and potential impacts of economic events or policy changes on stock prices.
3. **Timing Investments:** Market data analysis helps in timing investments by identifying favorable entry and exit points. This could involve using technical analysis (chart patterns, moving averages) or fundamental analysis (company earnings, growth prospects) to make decisions about buying or selling stocks.
4. **Long-Term Planning:** For long-term investors, analyzing market data provides insights into macroeconomic trends, demographic shifts, technological advancements, and other factors that could impact industries and companies over time. This foresight helps in building a resilient investment strategy that can withstand market cycles and economic changes.

In summary, analyzing share market data enhances decision-making capabilities by providing insights into market trends, risks, valuation metrics, and investment opportunities. It enables investors to formulate well-informed strategies for achieving their financial objectives and managing their portfolios effectively over time.

Aim of The Project :-

Share price of IT companies plays a significant role for investors and economy of the nation. For this reason, share price of two Indian major IT companies are taken under observation and analyzed by Time Series Analysis for predicting the future share prices of the companies. Using advanced statistical techniques, machine learning algorithms and comparing their share price on different time point in between the year 2021 to 2025, we seek to develop a reliable analysis and forecasting model which can be used by the investors, traders and researchers to take decision and manage risk efficiency. By that we can automatically understand in which company we should invest in near future. This project seeks to leverage time series analysis to contribute to the growing field of knowledge surrounding by share price of the IT companies and also will provide valuable insights into the future share market world.

Methodology

Here the methodology in which the project work is progressed is written below,

- First, I collected the stock price data on each starting trading day of every months from April 2021 to March 2025 of two major IT companies of INDIA (INFOSYS and HCLTECH) from a website and arranged the data in an excel file.
- Then using programming languages like R, Python, I did exploratory data analysis to visualize the data and to interpret the hidden patterns of the obtained dataset. We performed month-wise boxplots to study seasonal variation, plotted decomposed parts of the time series data to study the effects of various components.
- Next, under the data analysis part, at first, I plotted ACF plot and PACF plot to understand which model would be best fitted according to the obtained dataset and also checked stationarity using ADF test.
- Afterwards, I checked the normality of the residuals obtained from the fitted model using Shapiro-Wilk test. Then we obtained fitted values and forecasted values from the model we built and plotted those values w.r.t original stock price values to see how good our model can predict the fitted values. Also, we obtained several accuracy metrics for time series modeled data such as AIC, MAPE.

Detailed theory discussion of the statistical terms used in the project :-

Autoregressive [AR(1)] Model: A type of time series model where the current value of a series, Y_t, is expressed as a linear function of its past values. An AR model of order '1' is given by,

$$\mathbf{Y_t = c + \phi_1 Y_{t-1} + e_t}$$

- c and $\boldsymbol{\phi_1}$ are constants.
- e_t is a white noise error term (with zero mean and constant variance).
- The order of the AR model is 1.
- The model is stationary if $\mathbf{|\phi_1| < 1}$

Autoregressive [AR(2)] Model: A type of time series model where the current value of a series, Y_t, is expressed as a linear function of its past values and an error term. An AR model of order '2' is given by,

$$\mathbf{Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + e_t}$$

- $\boldsymbol{\phi_1}$ and $\boldsymbol{\phi_2}$ are constants.
- e_t is a white noise error term (with zero mean and constant variance).
- The order of the AR model is 2.
- The **stationarity** of an AR (2) model depends on the values of $\phi_1$and ϕ_2. The series is *stationary* if the roots of the characteristic equation
  $1 - \phi_1 z - \phi_2 z^2 = 0$, lie **outside the unit circle**.

Moving Average (MA) Model :- It is a type of stationary time series model where the current value of the series depends linearly on current and past random error terms. It is denoted as **MA(q)**, where *q* is the order of the model.

The general form of an MA(q) process is:

$$Y_t = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \cdots + \theta_q e_{t-q}$$

where:

- Y_t= value of the time series at time t,
- e_t= random white noise with mean 0 and variance σ_e^2,
- θ_i= parameters representing the weights of past error terms.

The **MA process** captures short-term correlations between observations but lacks long-term memory. Unlike autoregressive (AR) models, where dependence is on past values of the series, MA models depend on past sudden **shocks** (errors).

Stationarity of MA Process:

All **MA(q)** processes are *inherently stationary* because they are defined as a finite linear combination of white noise terms, which themselves are stationary.
Therefore, the mean, variance, and autocovariance of an MA process remain constant over time.

Autocorrelation Function (ACF): The ACF measures the correlation between a time series and its own lagged values. For a stationary AR(p) process, the ACF typically shows an exponential decay(**tails off**). For a MA(q) process, the ACF typically shows a **sharp cutoff to zero** after lag q. For a non-stationary AR(p) process, the ACF typically shows a **slow decay in magnitude,** as lag k increases, gradually approaching to zero. It helps in identifying the presence of the AR or MA components in the time series process.

Partial Autocorrelation Function (PACF): The PACF measures the correlation between a time series Y_t and its lag Y_{t-k} after removing the linear effects of the intermediate variables (Y_{t-1}, Y_{t-2}, ..., Y_{t-k+1}). For a MA(q) process, the PACF typically shows an exponential decay to zero (**tails off**). For a stationary AR(p) process, the ACF typically shows a **sharp cutoff to zero** after lag p. For a non-stationary AR(p) process, the ACF shows a very high spike at first lag(often close to 1) and then **cutoff and approaching to zero but slowly**. It helps in identifying the presence of the AR or MA components in the time series process.

Augmented Dickey Fuller Test: A statistical test used to determine whether a given time series process is stationary or not. To test the unit root hypothesis:

H_o : The process is non-stationary. vs **H_1 :** The process is stationary.

If the p-value is < alpha = 0.05, then at 5% level of significance, we reject the null hypothesis and conclude that the data is stationary.

Shapiro Wilk Test : A statistical test used to determine whether a the obtained residuals follow the normality assumption or not. To test the hypothesis:

H_o : residuals follow normality vs **H_1 :** residuals aren't normally distributed

If the p-value is > alpha = 0.05, then at 5% level of significance, we fail to reject the null hypothesis and conclude that the obtained residuals follow the normality assumption.
Hence, we can conclude that the fitted model is applicable in practical field.

Dataset

Timeline	Name of Companies	
	INFOSYS	HCLTECH
1st April 2021	1346.76	898.95
1st May 2021	1385.94	945.2
1st June 2021	1571.94	983.5
1st July 2021	1601.47	1024.95
1st August 2021	169689	1182.30
1st Sept 2021	1665.81	1279.55
1st Oct 2021	1658.40	1143.75
1st Nov 2021	1703.05	1138.65
1st Dec 2021	1877.17	1319.10
1st Jan 2022	1726.47	1099.40
1st Feb 2022	1705.99	1126.95
1st March 2022	189616	1163.75
1st April 2022	1558.77	1079.25
1st May 2022	1495.17	1040.75
1st June 2022	1453.71	973.25
1st July 2022	1541.02	948.4
1st August 2022	1484.58	938.95
1st Sept 2022	1405.53	932.35

1st Oct 2022	1529.03	1041.15
1st Nov 2022	1625.79	1120.80
1st Dec 2022	1499.75	1039.30
1st Jan 2023	1525.16	1122.20
1st Feb 2023	1479.21	1077.60
1st March 2023	1419.95	1085.25
1st April 2023	1245.73	1064.10
1st May 2023	1310.91	1145.10
1st June 2023	1328.02	1187.95
1st July 2023	1348.10	1116.70
1st August 2023	1427.41	1172.15
1st Sept 2023	1427.41	1234.80
1st Oct 2023	1360.73	1276.00
1st Nov 2023	1446.99	1340.90
1st Dec 2023	1534.25	1466.10
1st Jan 2024	1651.59	1576.00
1st Feb 2024	1664.52	1663.85
1st March 2024	1489.66	1543.55
1st April 2024	1412.59	1366.60
1st May 2024	1406.90	1324.10
1st June 2024	1566.75	1459.60
1st July 2024	1868.25	1642.60
1st August 2024	1943.70	1753.25

1st Sept 2024	1875.60	1796.10
1st Oct 2024	1757.25	1760.14
1st Nov 2024	1857.85	1841.86
1st Dec 2024	1880.00	1910.98
1st Jan 2025	1879.80	1725.45
1st Feb 2025	1687.70	1575.05
1st March 2025	1570.65	1592.50

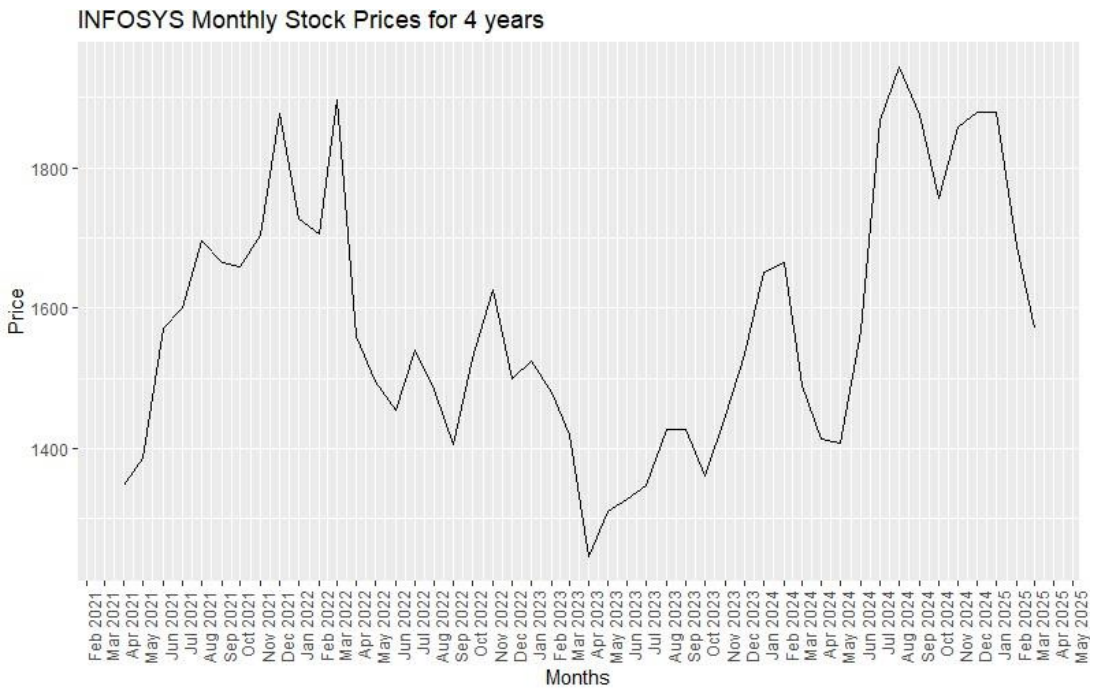
Exploratory Data Analysis

Data visualization of share price of IT companies over time provides an insightful summery of the changes during the taken time period. By plotting the values on a graph against their corresponding months, we can gain a visual understanding of the share price point fluctuations and potential patterns. Also we will examine the time series components present in the data by decomposing the obtained dataset.

Visualization can help investors and traders to make mindful decisions. It helps understand price ranges for setting stop loss or take profit levels and gauge the overall market sentiment.

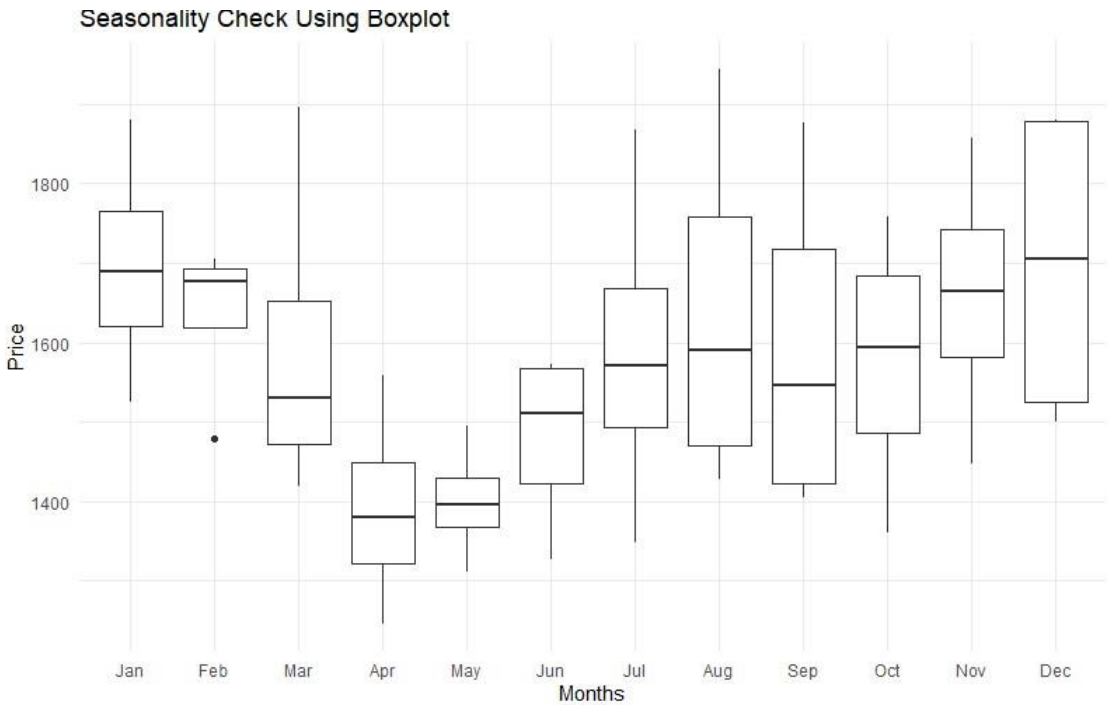
➢ For INFOSYS :-

a. Share Price Plot across months:-



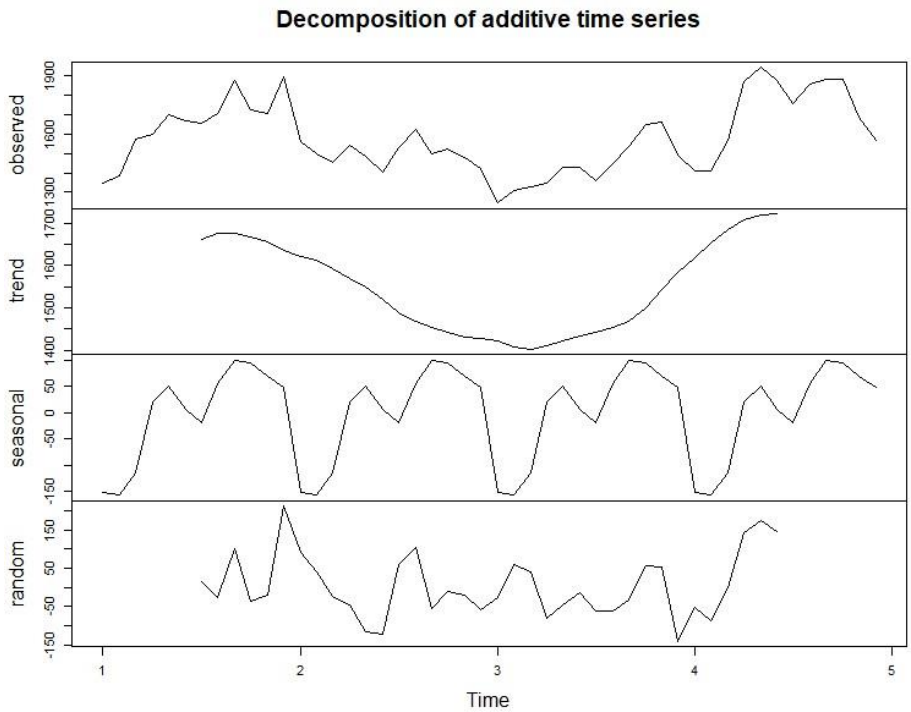
Interpretation :- From the above graph, we can conclude that there was an upward trend in the data at first; although at a middle point it showed a downward trend but eventually the company recovered from that phase and by the end of the time period, it again exhibited an upward trend. Also, the graph indicates that there may be some seasonal variation present in the data. Now, we will examine the presence of seasonal effect by drawing boxplots for each of the 12 months.

b. Seasonality Checking :-



Interpretation :- From the above boxplot, we can see that there is some seasonal variation present in the data. The median price is slightly on the higher side for the months January, February, November, December whereas that is on the lower side for the months April, May, June. There is no such effect of outlier.

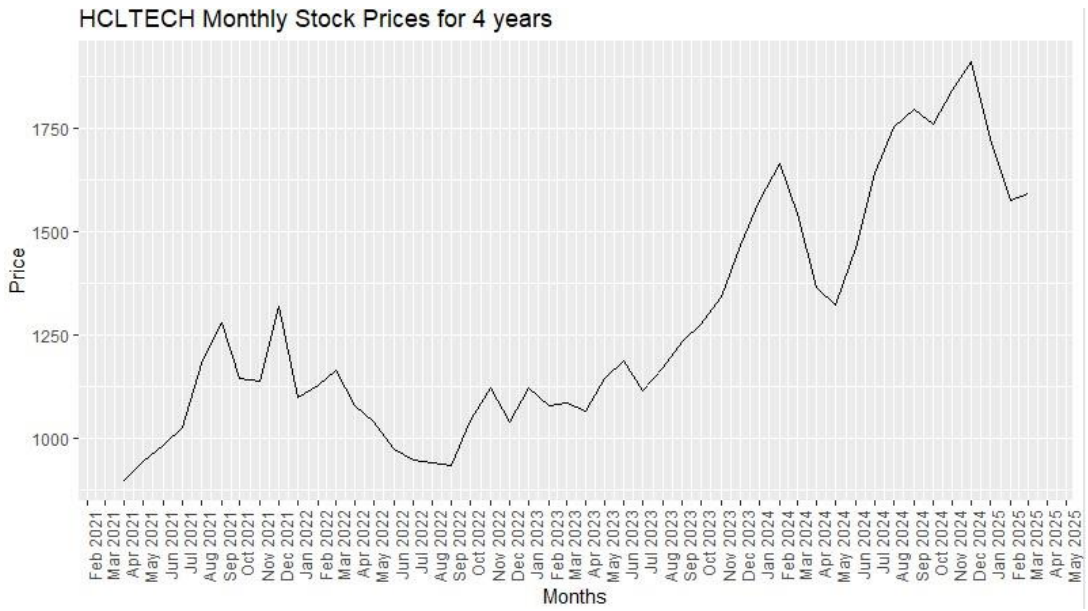
c. Decomposing the data :-



Interpretation :- Assuming the time series data to be of additive type (as magnitude of the seasonal fluctuations remains roughly constant over time), we got the different components shown in the above plot. We can see the trend can be modeled using a 2nd degree trend polynomial, there is some seasonal effect and also some random error components are present in the original data. The graph of the original data is shown in the top component of this plot. Adding one more thing to the analysis, decomposed trend component depicts that a 2nd degree trend polynomial will be a good fit but seeing the original data, one may conclude that trend curve would be of degree 3 or higher. The main reason behind this confusion is the decomposed trend graph is free from seasonal and random effects, whereas in the plot of the original data, it is a mixture of all the aforesaid three components, which arises the point of difference. Here, we should consider 2nd degree trend polynomial and the result of the effects of other components can be shown in the data.

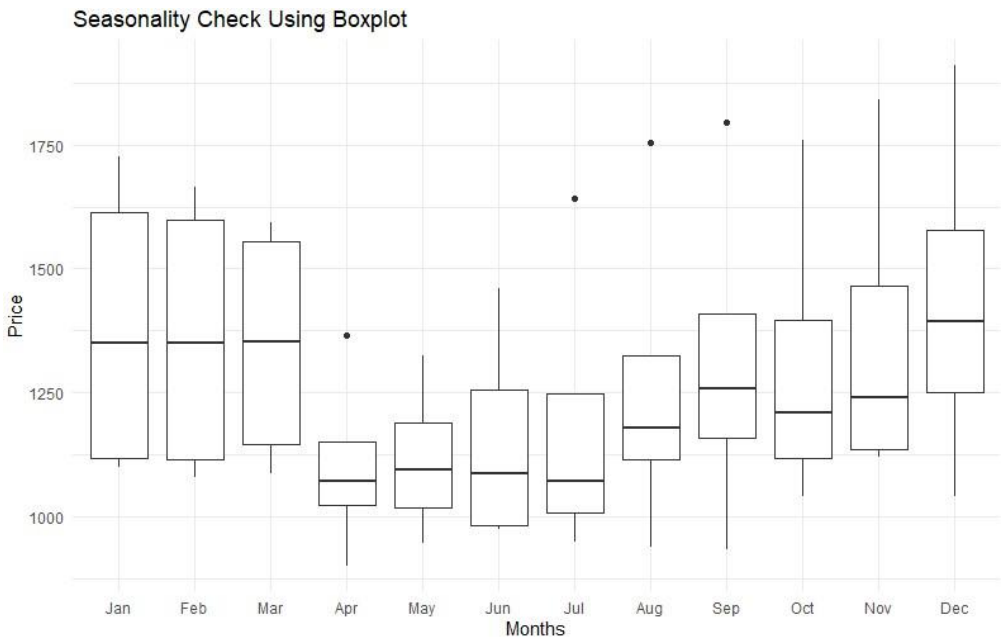
➢ For HCLTECH :-

d. Share Price Plot across months:-



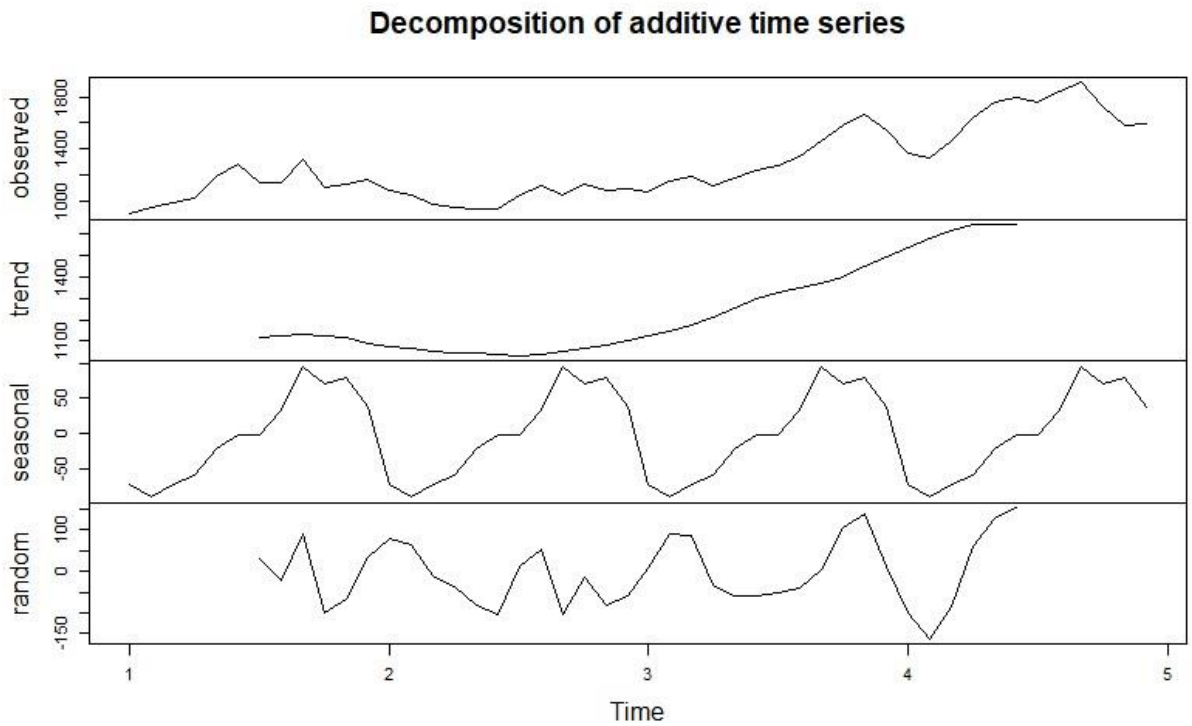
Interpretation :- From the above graph, we can conclude that there is an overall upward trend in the share market price throughout the 4 years for HCLTECH. Also, the graph indicates that there may be some seasonal variation present in the data. Now, we will examine the presence of seasonal effect by drawing boxplots for each of the 12 months.

e. Seasonality Checking :-



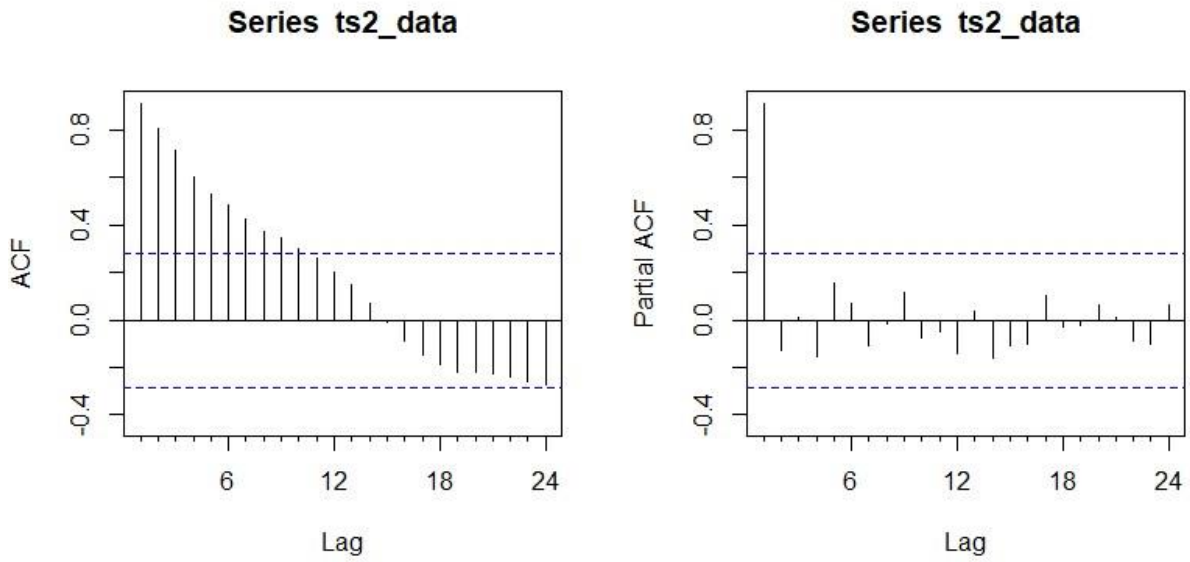
Interpretation :- From the above boxplot, we can see that there is some seasonal variation present in the data. The median price is slightly on the higher side for the months January, February, March, December whereas that is on the lower side for the months April, May, June, July. Also, there is no such effect of outliers in the dataset.

f. Decomposing the data :-



Interpretation :- Assuming the time series data to be of additive type (as magnitude of the seasonal fluctuations remains roughly constant over time), we got the different components shown in the above plot. We can see the trend can be modeled using a linear trend equation, there is some seasonal effect and also some random error components are present in the original data. The graph of the original data is shown in the top component of this plot.

g. ACF & PACF plot :-



Interpretation :- From the above ACF & PACF plots of the stock price data of HCLTECH, we can conclude that the original data is non-stationary and it can be modeled using AR(1) model as in the PACF plot, there is a high spike at lag 1, after that the plot shows a sharp cutoff and goes to zero rapidly. Also, it indicates that some seasonal fluctuation may be present in the data.

Data Analysis

In this part, we will analyze the data of different companies by checking the stationarity of the obtained time series data, will see the nature of the ACF & PACF plots and fit suitable models accordingly, will check the normality assumptions of the residuals, accuracy of the models and afterwards we will try to get the fitted values, compare with the actual dataset and will try to forecast the next possible price values using the model.

- **Analysis for INFOSYS :-**

  To check the stationarity, we performed Augmented Dickey Fuller Test on the obtained dataset and found the following,

  Augmented Dickey-Fuller Test

  data: ts_data
  Dickey-Fuller = -1.9444, Lag order = 3, p-value = 0.5963
  alternative hypothesis: stationary

  As the p-value is greater than 0.5, hence at 5% level of significance, we are unable to reject the null hypothesis of the data being non-stationary and finally conclude that the original data is non-stationary. Though this result is very common in case of stock price data. Now after differencing the data once and again performing ADF Test on the differenced data, we got the following,

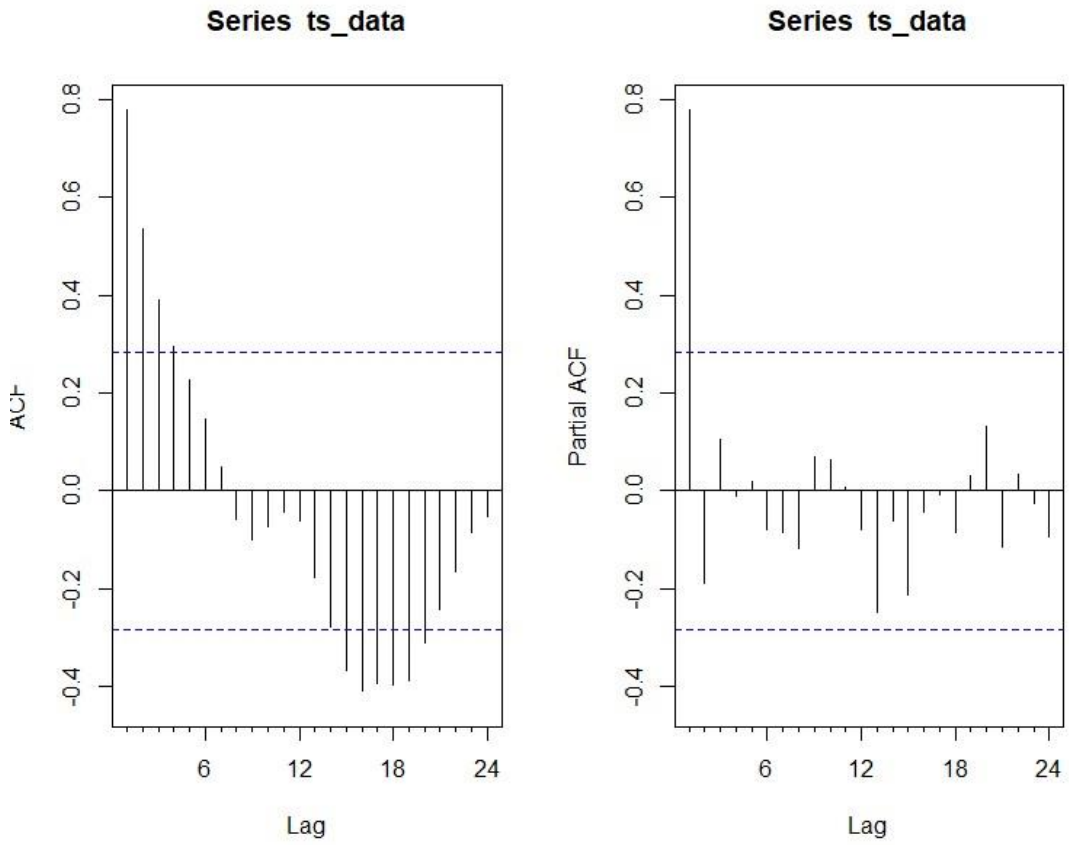
  Augmented Dickey-Fuller Test

  data: diff_data
  Dickey-Fuller = -3.8911, Lag order = 3, p-value = 0.02223
  alternative hypothesis: stationary

  So here, p-value is lesser than 0.05 and hence we say that at 5% level of significance, we are able to reject the null hypothesis of the data being non-stationary and hence the data is stationary after performing differencing once.

  Now, we will see the ACF plot and PACF plot of the data to determine the order of the model, we are going to fit.

a. ACF & PACF plots :-



Interpretation :- The ACF plot shows a slow decay as lag increases, which indicates that the data is non-stationary at the beginning, confirming the results of the ADF test. On the other hand, the PACF plot shows a sharp cutoff after lag 1 and eventually nearing zero for all subsequent lags, indicating the presence of AR(1) component and no MA component. So, we will fit an ARIMA (1,1,0) model to this data.

Model Fitting :- By fitting ARIMA(1,1,0) model using machine learning techniques, we got the following,

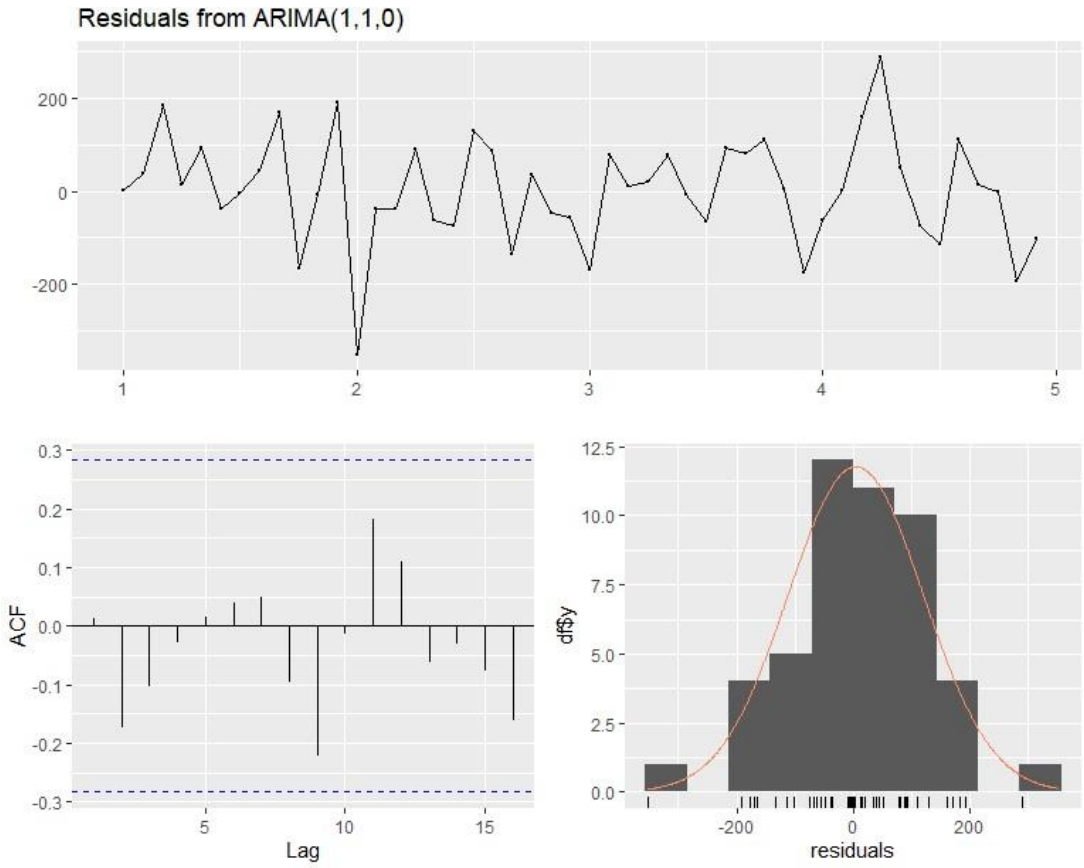
```
Call:
arima(x = ts_data, order = c(1, 1, 0))

Coefficients:
         ar1
      0.0796
s.e.  0.1456

sigma^2 estimated as 13439:  log likelihood = -290.08,  aic = 582.16
```

We obtained the fitted values, residuals using this model and checked the normality assumption of the residuals, plotted the fitted values with the original values, which are shown below,

b. Normality Check of Residuals :-



Interpretation :- Seeing the above plot, we can conclude that the residuals of the fitted ARIMA(1,1,0) model approximately follows normality. To check it further, we performed Shapiro-Wilk test and got the results,

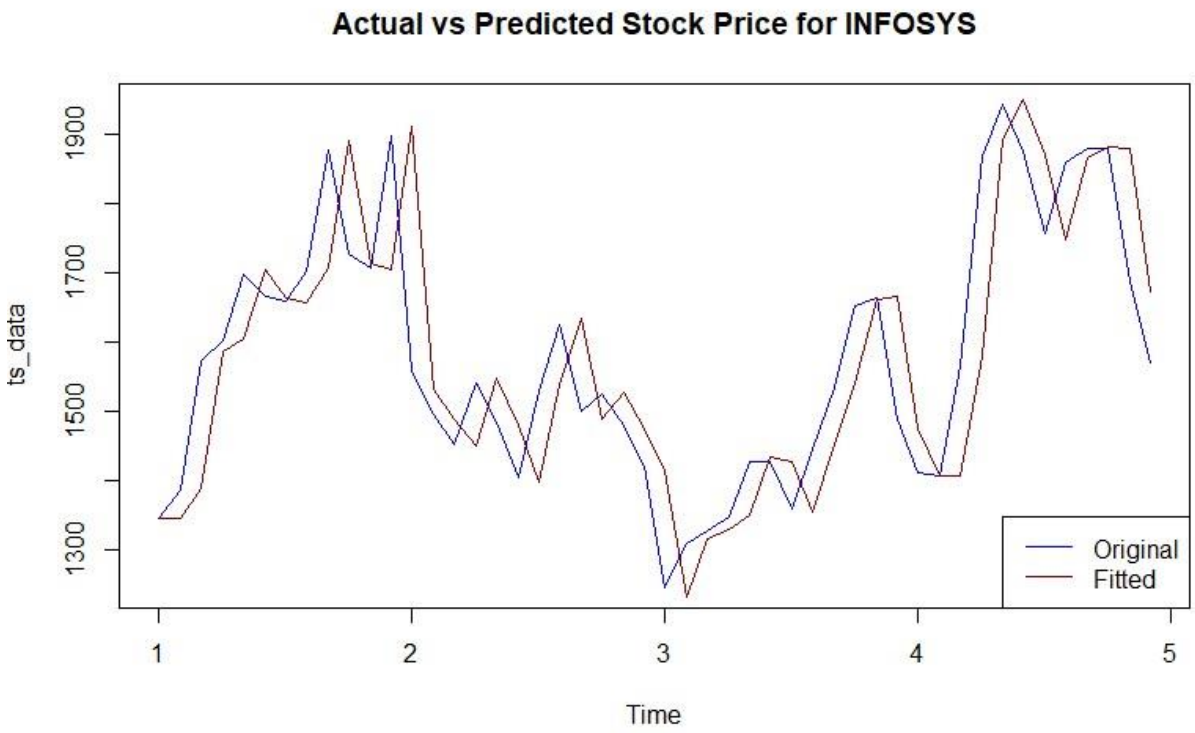
Shapiro-Wilk normality test

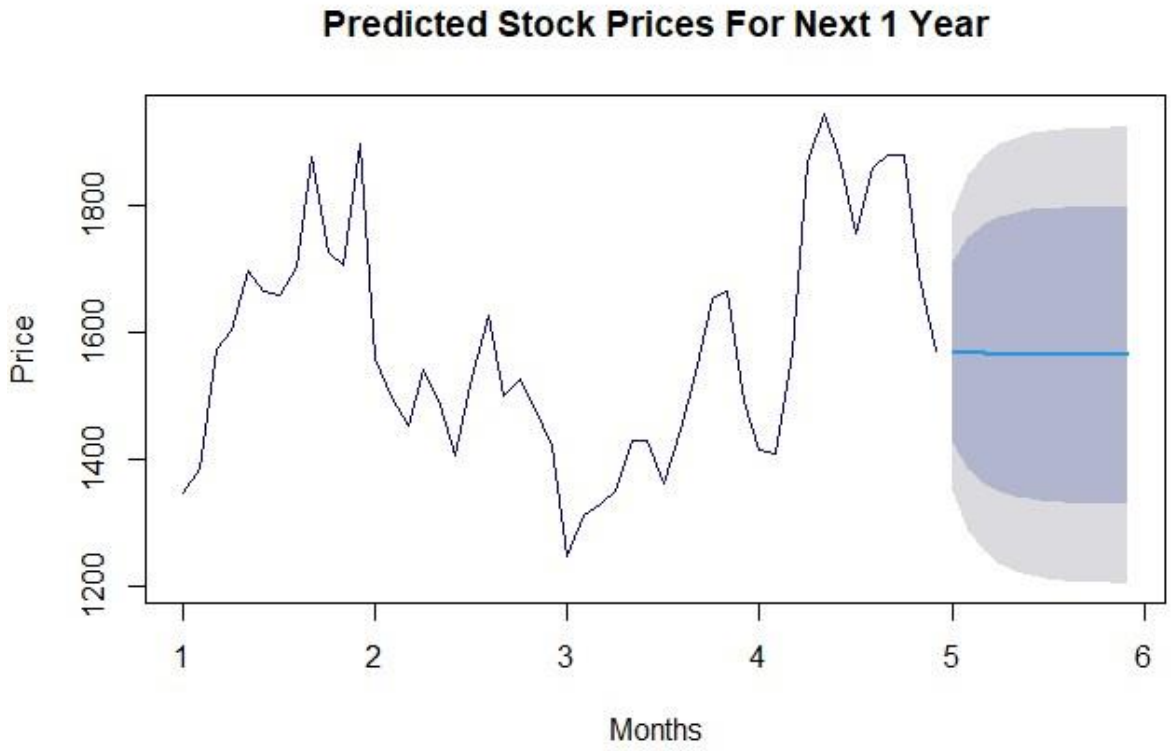
data: residuals

W = 0.98157, p-value = 0.6452

Hence, at 5% level of significance, our null hypothesis of following normality gets accepted and finally we conclude that the residuals of the fitted model are normally distributed.

c. Plots showing fitted values and forecasted values :-

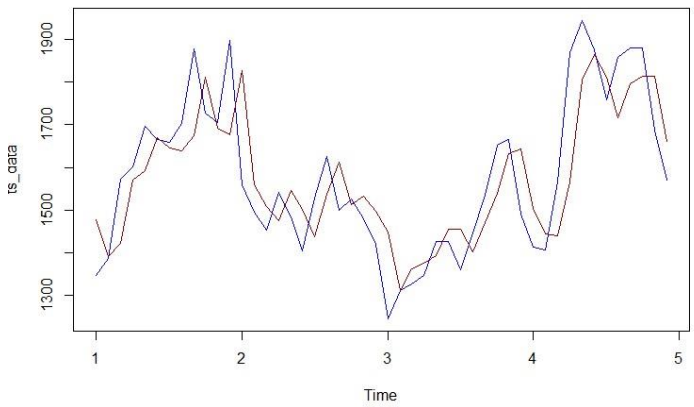




Interpretation :- We can conclude from the fitted values vs observed values curve, that the fitted model is a good fit for the observed data and in the next plot, we tried to show the forecasted stock prices of the company INFOSYS for the next one year.

Note :-

By using the machine learning algorithm via programming language, we got that the best fitted model for this data will be ARIMA (0,1,0), that means there will be no AR and no MA components, just one order differencing component will be present. But, upon seeing the ACF and PACF plot, we can say that AR(1) component is present in the data. So, I trusted upon my intuition and used the previously fitted ARIMA(1,1,0) model to fit the obtained data. Also, from the plot given here, we can see that our previously used model gave better fitted values w.r.t the observed original data. Also that model had lesser accuracy metrics such as lesser AIC and lesser MAPE (5.46 in case of this model and 5.45 in case of previously used model). Hence, finally we chose that ARIMA (1,1,0) model for our data.



- **Analysis for HCLTECH :-**

To check the stationarity, we performed Augmented Dickey Fuller Test on the obtained dataset and found the following,

Augmented Dickey-Fuller Test

data: ts2_data
Dickey-Fuller = -2.2577, Lag order = 3, p-value = 0.4712
alternative hypothesis: stationary

As the p-value is greater than 0.5, hence at 5% level of significance, we are unable to reject the null hypothesis of the data being non-stationary and finally conclude that the original data is non-stationary. Though this result is very common in case of stock price data. Now after differencing the data once and again performing ADF Test on the differenced data, we got the following,

Augmented Dickey-Fuller Test

data: diff2_data
Dickey-Fuller = -3.9332, Lag order = 3, p-value = 0.02049
alternative hypothesis: stationary

So here, p-value is lesser than 0.05 and hence we say that at 5% level of significance, we are able to reject the null hypothesis of the data being non-stationary and hence the data is stationary after performing differencing once.

Model Fitting :- Using R programming language, we obtain the followings,

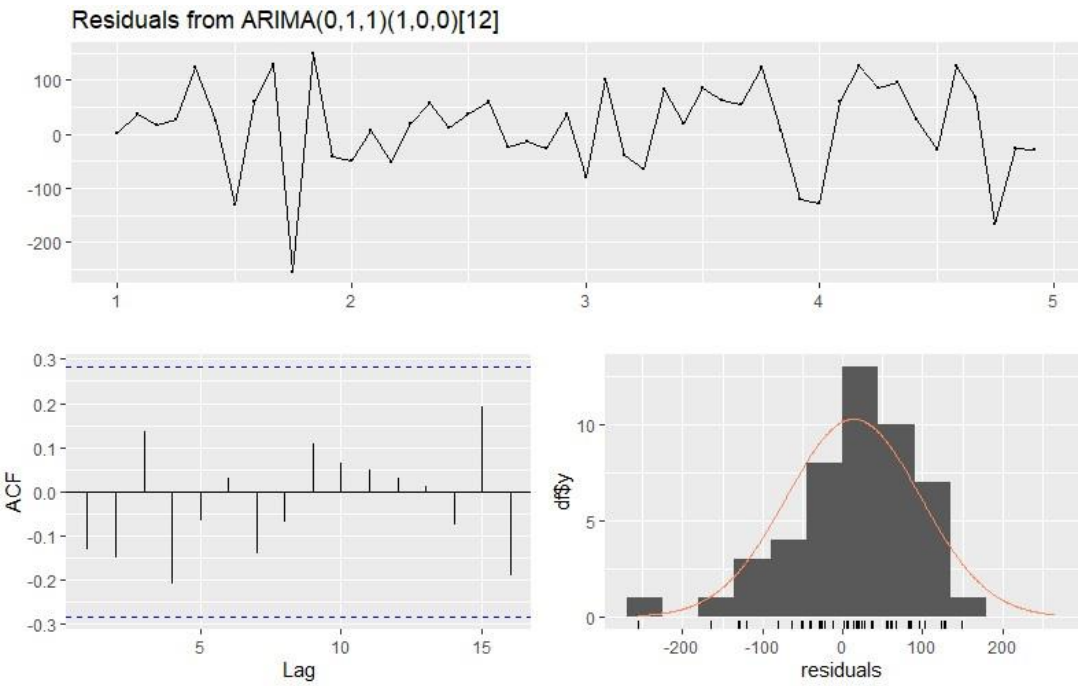
```
Series: ts2_data
ARIMA(0,1,1)(1,0,0)[12]

Coefficients:
         ma1     sar1
      0.4903  -0.4848
s.e.  0.1992   0.1697

sigma^2 = 7515:  log likelihood = -277.14
AIC=560.28   AICc=560.84   BIC=565.83
```

We can see that the best fitted model is a SARIMA model accounting both seasonal and non-seasonal parts of the time series data. Here, we have seasonal AR(1) with period 12, which captures yearly pattern(monthly data) and in the non-seasonal part, we have d=1 and order of best fitted MA component is 1. We obtained the fitted values, residuals using this model and checked the normality assumption of the residuals, plotted the fitted values with the original values, which are shown below,

d. Normality Check of Residuals :-



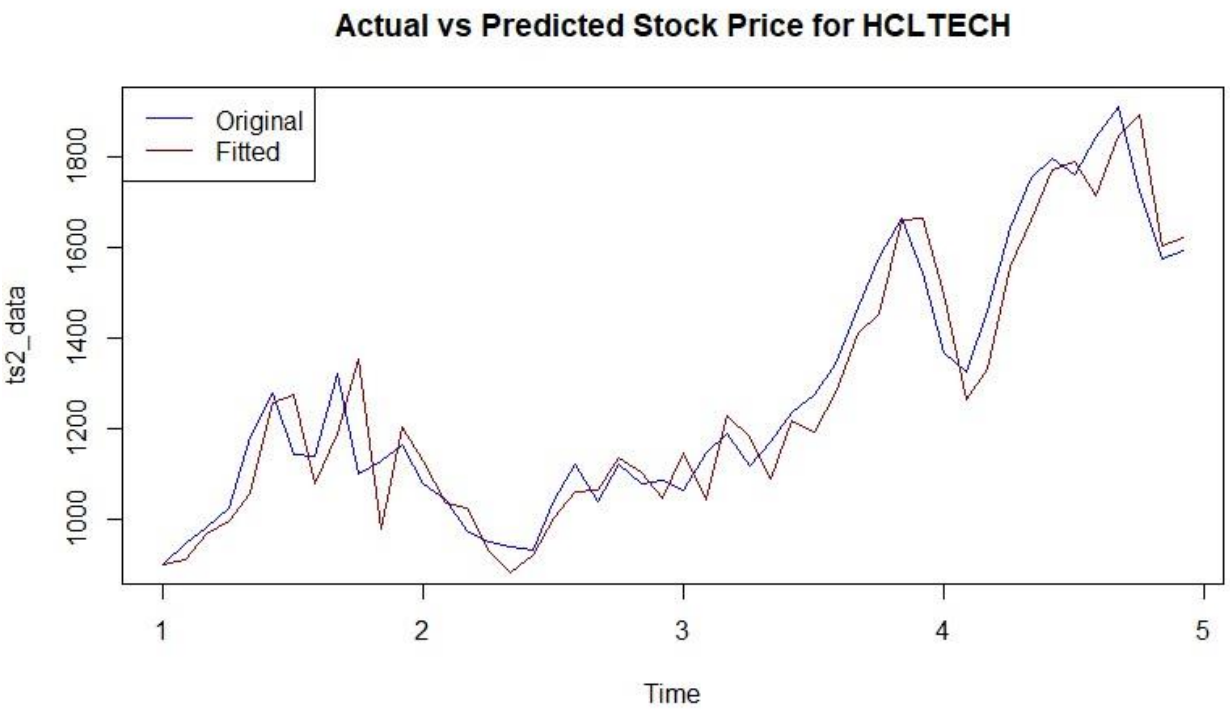
Interpretation :- Seeing the above plot, we can conclude that the residuals of the fitted SARIMA(0,1,1)(1,0,0)[12] model approximately follows normality. To check it further, we performed Shapiro-Wilk test and got the results,

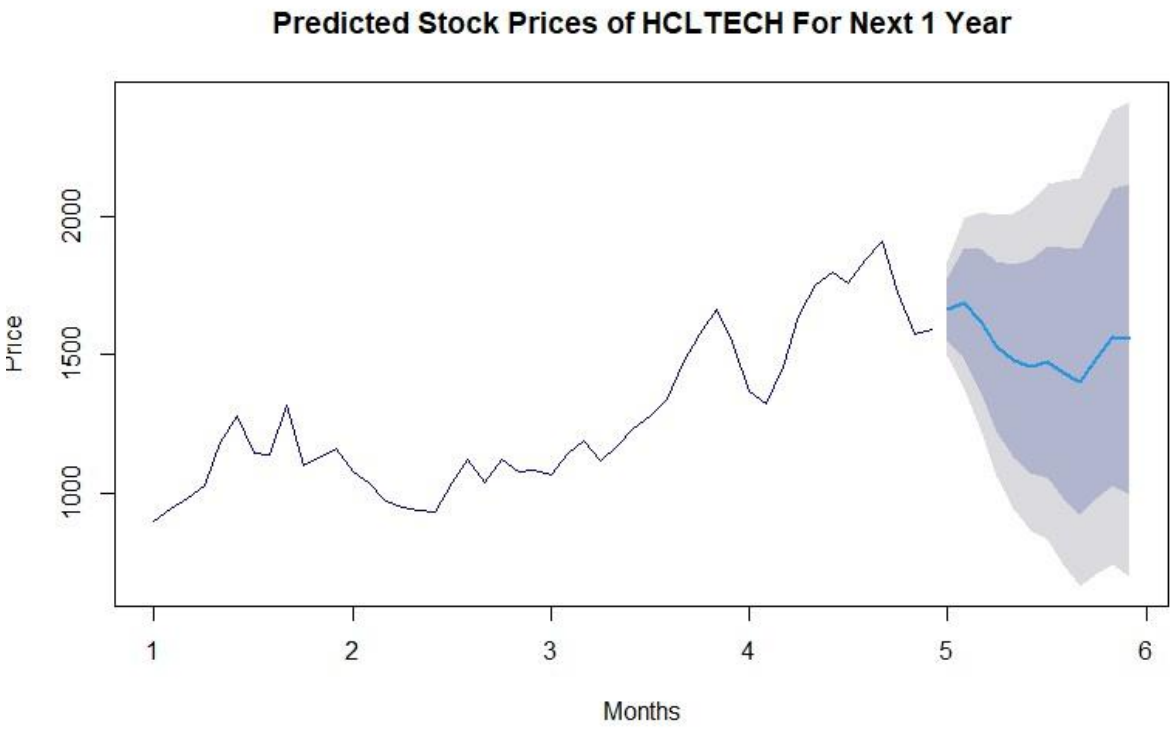
Shapiro-Wilk normality test

data: residuals
W = 0.95107, p-value = 0.6413

Hence, at 5% level of significance, our null hypothesis of following normality gets accepted and finally we conclude that the residuals of the fitted model are normally distributed.

e. Plots showing fitted values and forecasted values :-





Interpretation :- We can conclude from the fitted values vs observed values curve, that the fitted model is a good fit for the observed data and in the next plot, we tried to show the forecasted stock prices of the company HCLTECH for the next one year. Also, we got a lower MAPE value (5.21) for fitted model.

Final Conclusion

Here, upon doing this project works, finally we came to the following conclusions,

1. For both the companies INFOSYS & HCLTECH, the stock price data at the starting point of each months on a span of 4 years (from April 2021 to March 2025), is a non-stationary time series data.
2. Using differencing of appropriate order, we made two time series data stationary and built time series model accordingly. For HCLTECH, we fitted a SARIMA(0,1,1)(1,0,0)[12] model and for INFOSYS, we fitted a ARIMA(1,1,0) model.
3. By doing the data visualization, we tried to show the components of a time series in the obtained datasets and drew conclusions accordingly.
4. From the obtained models, we obtained residuals, checked normality assumption and upon verification, we further proceeded to find fitted and forecasted values. Both type of values for both the companies are shown in the above works. Also, we obtained MAPE & AIC values for obtained models to check the accuracy level of our time series models.

References

The sources from where I gathered knowledge to make this project are :

I. HCLTECH–Aug 25

⇨ Investing.com

⇨ Link : https://in.investing.com/equities/hcl-technologies-historical-data?cid=1205457&end_date=1714291190&st_date=1522261800&interval_sec=monthly

2. Applied statistics by PARIMAL MUKHOPADHYAY

3. Fundamentals Of Statistics, Volume II , A.M GOON, M.K. DASGUPTA , B. DASGUPTA

4. A special thanks to Microsoft excel and R studios, Python for making the project works much much easier.

Used Codes

```
> library(ggplot2)
> library(forecast)
> library(lubridate)
> library(readxl)
> library(TSA)
> library(tseries)
> getwd()
[1] "C:/Users/Tatai/Documents"
> df<-read_excel("C:/Users/Tatai/Downloads/INFOSYS.xlsx")
New names:
• `` -> `...8`
• `` -> `...9`
• `` -> `...10`
> df<-df[,-c(8,9,10)]
> df$Date<-as.Date(df$Date)
> head(df)
# A tibble: 6 × 7
  Date       Price  Open  High   Low Vol.    `Change %`
  <date>     <dbl> <dbl> <dbl> <dbl> <chr>        <dbl>
1 2021-04-01 1347. 1372. 1469. 1313. 174.38M    -0.01
2 2021-05-01 1386. 1333. 1408. 1304. 110.64M     0.0291
3 2021-06-01 1572. 1392. 1582. 1357. 147.30M     0.134
4 2021-07-01 1601. 1568. 1614. 1525. 109.51M     0.0188
5 2021-08-01 1697. 1618. 1747. 1610. 132.75M     0.0596
6 2021-09-01 1666. 1700. 1778. 1646. 113.96M    -0.0183
> par(mfrow=c(1,1))
> ggplot(df, aes(x = Date, y = Price)) +
+     geom_line() +
+     scale_x_date(date_labels = "%b %Y",   # Jan 2021, Feb 2021...
+                  date_breaks = "1 month") +
+     theme(axis.text.x = element_text(angle = 90, hjust = 1)) + labs(title="INFOSYS
Monthly Stock Prices for 4 years",x="Months",y="Price")
> df$Month <- month(df$Date, label = TRUE)
> df$Month <- factor(df$Month, levels = month.abb)
> ggplot(df, aes(Month, Price)) +
+     geom_boxplot() +
+     labs(
+       title = "Seasonality Check Using Boxplot",
+       x = "Months",
+       y = "Price"
+     ) +
+     theme_minimal()
> decomposed_data<-decompose(ts_data,type="additive")
> plot(decomposed_data)
```

```
> ts_data<-ts(df$Price,frequency=12)
> par(mfrow=c(1,2))
> Acf(ts_data)
> Pacf(ts_data)
> print(adf_results<-adf.test(ts_data))

        Augmented Dickey-Fuller Test

data:  ts_data
Dickey-Fuller = -1.9444, Lag order = 3, p-value = 0.5963
alternative hypothesis: stationary
> diff_data<-diff(ts_data)
> adf.test(diff_data)

        Augmented Dickey-Fuller Test

data:  diff_data
Dickey-Fuller = -3.8911, Lag order = 3, p-value =
0.02223
alternative hypothesis: stationary
> model<-arima(ts_data,order=c(1,1,0))
> model

Call:
arima(x = ts_data, order = c(1, 1, 0))

Coefficients:
        ar1
     0.0796
s.e.  0.1456

sigma^2 estimated as 13439:  log likelihood = -290.08,  aic = 582.16
> residuals<-residuals(model)
> print(shapiro_test<-shapiro.test(residuals))

        Shapiro-Wilk normality test

data:  residuals
W = 0.98157, p-value = 0.6452
> fitted_values<-fitted(model)
> fitted_ts<-ts(fitted_values,start=start(ts_data),frequency=12)
> par(mfrow=c(1,1))
> plot(ts_data,col="blue",type="l")
> lines(fitted_ts,col="darkred")
> checkresiduals(model)

        Ljung-Box test
```

```
data:  Residuals from ARIMA(1,1,0)
Q* = 6.0436, df = 9, p-value = 0.7355

Model df: 1.   Total lags used: 10
> print(best_model<-auto.arima(ts_data))
Series: ts_data
ARIMA(1,0,0) with non-zero mean

Coefficients:
         ar1      mean
      0.7920  1562.475
s.e.  0.0853    71.258

sigma^2 = 12604:  log likelihood = -294.18
AIC=594.37   AICc=594.91   BIC=599.98
> residuals<-residuals(best_model)
> print(shapiro_test<-shapiro.test(residuals))

        Shapiro-Wilk normality test

data:  residuals
W = 0.99218, p-value = 0.986
> fitted_values<-fitted(best_model)
> checkresiduals(best_model)

        Ljung-Box test

data:  Residuals from ARIMA(1,0,0) with non-zero mean
Q* = 5.2945, df = 9, p-value = 0.8079

Model df: 1.   Total lags used: 10
> fitted_ts<-ts(fitted_values,start=start(ts_data),frequency=12)
> plot(ts_data,col="blue",type="l",main="Actual vs Predicted Stock Price for
INFOSYS")
> lines(fitted_ts,col="darkred")
>
legend("bottomright",lty=c(1,1),col=c("blue","darkred"),legend=c("Original","Fitted")
)
> forecast_values<-forecast(model,h=12)
> plot(forecast_values,main="Predicted Stock Prices For Next 1
Year",ylab="Price",xlab="Months",col="navyblue")
> mape<-function(actual,predicted){
+ mean(abs((actual-predicted)/actual))*100}
> print(mape1_value<-mape(df$Price,fitted1_values))
[1] 5.452686
> print(mape1_value<-mape(df$Price,fitted_values))
[1] 5.449381
```

```
> df2<-read_excel("C:/Users/Tatai/Downloads/HCLT.xlsx")
New names:
• `` -> `...8`
• `` -> `...9`
• `` -> `...10`
> df2<-df2[,-c(8,9,10)]
> df2$Date<-as.Date(df2$Date)
> head(df2)
# A tibble: 6 × 7
  Date       Price  Open  High   Low Vol.    `Change %`
  <date>     <dbl> <dbl> <dbl> <dbl> <chr>        <dbl>
1 2021-04-01  899.  981. 1043.  895. 136.31M    -0.0752
2 2021-05-01  945.  899.  957.  891  101.42M     0.0514
3 2021-06-01  984.  948  1010   935. 78.20M      0.0405
4 2021-07-01 1025.  987  1052.  960. 97.81M      0.0421
5 2021-08-01 1182. 1034  1193. 1023. 110.95M     0.154
6 2021-09-01 1280. 1190  1378. 1154  102.24M     0.0823
> par(mfrow=c(1,2))
> ts2_data<-ts(df2$Price,frequency=12)
> Acf(ts2_data)
> Pacf(ts2_data)
> par(mfrow=c(1,1))
> ggplot(df2, aes(x = Date, y = Price)) +
+ geom_line() +
+ scale_x_date(date_labels = "%b %Y",   # Jan 2021, Feb 2021...
+       date_breaks = "1 month") +
+ theme(axis.text.x = element_text(angle = 90, hjust = 1)) + labs(title="HCLTECH
Monthly Stock Prices for 4 years",x="Months",y="Price")
> df2$Month <- month(df2$Date, label = TRUE)
> df2$Month <- factor(df2$Month, levels = month.abb)
> ggplot(df2, aes(Month, Price)) +
+     geom_boxplot() +
+     labs(
+     title = "Seasonality Check Using Boxplot",x = "Months",y = "Price") +
theme_minimal()
> decomposed2_data<-decompose(ts2_data,type="additive")
> plot(decomposed2_data)
> print(adf_result2<-adf.test(ts2_data))

        Augmented Dickey-Fuller Test

data:  ts2_data
Dickey-Fuller = -2.2577, Lag order = 3, p-value =
0.4712
alternative hypothesis: stationary
> diff2_data<-diff(ts2_data)
> print(adf_result2<-adf.test(diff2_data))
```

```
        Augmented Dickey-Fuller Test

data:  diff2_data
Dickey-Fuller = -3.9332, Lag order = 3, p-value =
0.02049
alternative hypothesis: stationary
> print(best2_model<-auto.arima(ts2_data))
Series: ts2_data
ARIMA(0,1,1)(1,0,0)[12]

Coefficients:
         ma1     sar1
      0.4903  -0.4848
s.e.  0.1992   0.1697

sigma^2 = 7515:  log likelihood = -277.14
AIC=560.28   AICc=560.84   BIC=565.83
> fitted2_values<-fitted(best2_model)
> fitted2_ts<-ts(fitted2_values,start=start(ts2_data),frequency=12)
> forecast2_values<-forecast(best2_model,h=12)
> plot(forecast2_values,main="Predicted Stock Prices of HCLTECH For Next 1
Year",ylab="Price",xlab="Months",col="navyblue")
> plot(ts2_data,col="blue",type="l",main="Actual vs Predicted Stock Price for
HCLTECH")
> lines(fitted2_ts,col="darkred")
> legend("topleft",lty=c(1,1),col=c("blue","darkred"),legend=c("Original","Fitted"))
> par(mfrow=c(1,2))
> checkresiduals(best2_model)
        Ljung-Box test
data:  Residuals from ARIMA(0,1,1)(1,0,0)[12]
Q* = 8.084, df = 8, p-value = 0.4253

Model df: 2.   Total lags used: 10
> residuals<-residuals(best2_model)
> print(shapiro_test<-shapiro.test(residuals))

        Shapiro-Wilk normality test

data:  residuals
W = 0.95107, p-value = 0.6413
> mape<-function(actual,predicted){
+ mean(abs((actual-predicted)/actual))*100}
> print(mape2_value<-mape(df2$Price,fitted2_values))
[1] 5.209455
```